

Chapter 55: Thromboembolic Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2539–2583 of the PDF. High-yield numbers are in **bold**. The six tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 5 figures are included from the PDF.

Big picture: VTE is rare (**1–2 per 1000 pregnancies**) but **~5× the nonpregnant risk**, and PE causes **~10% of pregnancy-related deaths**. Pregnancy raises all three parts of Virchow's triad. The biggest risk factors are **prior VTE** and **thrombophilia**. Diagnosis starts with **compression ultrasound**; treatment is **LMWH** (warfarin only postpartum or with mechanical valves). Prophylaxis guidelines disagree widely.

1. Epidemiology

- Early ambulation dramatically ↓ puerperal VTE, yet it remains a **leading cause of maternal morbidity and mortality**.
- **Thrombotic PE ≈ 10% of US pregnancy-related deaths** (2011–2015).
- Incidence **1–2 per 1000 pregnancies**, **~5× nonpregnant**.
- Cases split ~equally antepartum and postpartum, but **DVT alone is more frequent antepartum**, **PE is more common postpartum**.
- Postpartum incidence: **22 per 100,000 deliveries in weeks 0–6**, **~3 per 100,000 in weeks 7–12**.
- **Postthrombotic syndrome** in up to **2%**.

2. Pathophysiology

- **Virchow's triad** (1856) — all ↑ in pregnancy:
 1. **Stasis** — uterus compresses pelvic veins and IVC; **leg venous flow velocity ↓ 50%** from the early 3rd trimester to **6 wk postpartum**. **Stasis = the most constant predisposing factor**.
 2. **Endothelial injury** — stasis, delivery, preeclampsia, sepsis.
 3. **Hypercoagulability** — ↑ synthesis of most clotting factors.
- **Most important risk factor = personal history of thrombosis: 15–25%** of pregnancy VTE are recurrences.
- **20–50%** of women with pregnancy/postpartum VTE have an identifiable **inherited thrombophilia**.
- Risk **~2×** with multifetal gestation, anemia, hyperemesis, hemorrhage, **cesarean**, obesity, preeclampsia, postpartum infection; higher after **stillbirth** or **peripartum hysterectomy**. **IBD 2–3×**.

Table 55-1. Some Risk Factors Associated with an Increased Risk for VTE

Obstetrical	General
Cesarean delivery	Age 35 years or older
Cesarean hysterectomy	Anatomical anomaly ^a
Diabetes mellitus	Antiphospholipid antibodies
Hemorrhage and anemia	Connective tissue disease
Hyperemesis	Dehydration
Immobility — prolonged bed rest	Hormonal contraceptives
Multifetal gestation	Hospitalization
Multiparity	Immobility
Preeclampsia	Infection and inflammatory disease
Puerperal infection	Inflammatory bowel disease
Stillbirth	Malignancy
	Nephrotic syndrome
	Obesity
	Orthopedic procedures
	Paraplegia
	Prior VTE
	Sickle-cell disease
	Smoking
	Surgery
	Thrombophilia

^aIncludes May-Thurner syndrome (iliac vein compression syndrome). VTE = venous thromboembolism.

3. Thrombophilias

- Inherited or acquired deficiencies of the **inhibitory (regulatory) proteins** of coagulation → hypercoagulability and recurrent VTE.
- Present in **~15%** of white Europeans but responsible for **~50%** of pregnancy VTE.

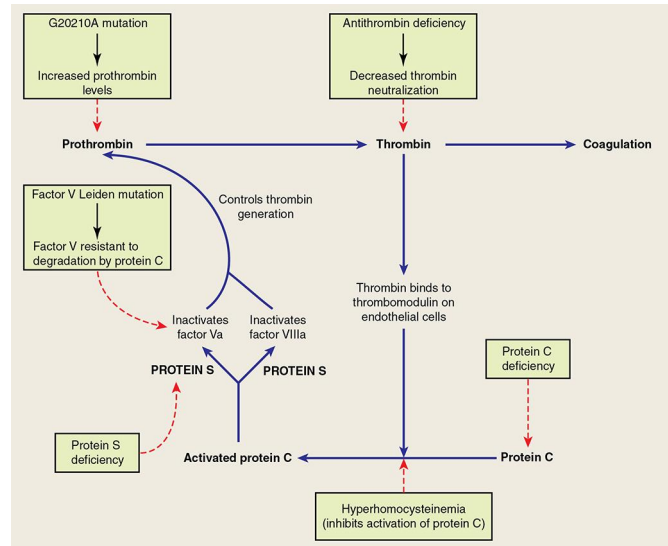


Figure 55-1. Where each thrombophilia acts: G20210A ↑ prothrombin; antithrombin deficiency ↓ thrombin neutralization; FVL resists protein C; protein C/S deficiency and hyperhomocysteinemia impair the protein C pathway that inactivates Va and VIIIa.

Table 55-2. Inherited Thrombophilias and Their Association with Venous Thromboembolism (VTE) in Pregnancy

Thrombophilia	VTE risk/pregnancy (%): No History	Prior VTE	Family History
Factor V Leiden heterozygote	0.5–3.1	10	0.06–1.2
Factor V Leiden homozygote	2.2–14	17	1–16
Prothrombin gene heterozygote	0.4–2.6	>10	0–0.73
Prothrombin gene homozygote	2–4	>17	1.6
FVL/prothrombin double heterozygote	8.2	>20	0–2.4
Antithrombin deficiency	0.2–1.16	40	0–8
Protein C deficiency	0.1–5.4	4–17	0–5
Protein S deficiency	0.1–6.6	0–22	0–1.5
Hyperhomocysteinemia	0	0	<1

Data from Abbattista, 2020; ACOG, 2020a; Bates, 2018; Croles, 2017. The printed FVL homozygote "no history" value reads "2.2–1.4" (reversed), probably 2.2–14. The printed family-history column conflicts with the text, which gives ≥10% for FVL heterozygotes (and >10% for prothrombin heterozygotes) with a personal or family history, and 6–7% for protein S deficiency with a family history.

Inherited thrombophilias — overview

- Often a **family history**; found in **up to ½ of patients with VTE before age 45**, especially unprovoked.
- Most significant family history: **sudden death from PE** or several relatives on long-term anticoagulation.

Antithrombin deficiency

- Liver-made; inactivates **thrombin and factor Xa**. **Heparin accelerates antithrombin >1000-fold**.
- Autosomal dominant. **Type I** ↓ synthesis (normal protein); **type II** normal levels, ↓ function.
- **Homozygous = lethal**. Heterozygous **~1 in 2500**.
- **⚠ The most thrombogenic heritable thrombophilia: 25–50× RR** of VTE in the general population; **6×** in pregnancy.
- 18 pregnancies: untreated → 3 VTE, **50% stillbirth and FGR**; LMWH-treated → no VTE or stillbirth (¼ FGR). VTE **7% with LMWH vs 12%** without.
- **Anticoagulate in pregnancy regardless of prior thrombosis**. When anticoagulation must be withheld (surgery, delivery) or VTE occurs on treatment → **recombinant human antithrombin**.

Protein C deficiency

- Thrombin + thrombomodulin activates **protein C**, which with **protein S** inactivates **Va and VIIIa**. Protein C activity rises modestly in the first half of pregnancy.
- >160 autosomal dominant mutations; heterozygous prevalence **2–3 per 1000** (variable expression); lab threshold **50–60%** activity.
- **6–12× VTE risk**. Homozygous → **fatal neonatal purpura fulminans**; sepsis can cause purpura fulminans in heterozygous adults.

Protein S deficiency

- Cofactor for activated protein C. >130 mutations; prevalence **0.3–1.3 per 1000**.
- **Free, functional and total protein S all fall in normal pregnancy** (and with some OCs) → diagnosis in pregnancy is difficult.
- If screening in pregnancy is needed: **free protein S antigen <30% (2nd trimester), <24% (3rd trimester)**.
- VTE risk ↑ severalfold; **6–7% with a positive family history**. Maternal **cerebral vein thrombosis** described (5 of 29 pregnant women had thrombosis). Neonatal homozygous → purpura fulminans.

Factor V Leiden (FVL)

- **Most prevalent thrombophilia** → **activated protein C resistance**. Missense mutation: **glutamine for arginine at position 506** → factor Va inactivated **~10× more slowly**.
- **Heterozygous FVL = most common heritable thrombophilia: 3–15%** of select Europeans, **1.2%** African Americans, **2.2%** Hispanic Americans; **virtually absent in African blacks and Asians**.
- Heterozygotes account for **~40% of pregnancy VTE**.

FVL genotype	No personal/family history	With personal or first-degree family history (<50 y)
Heterozygous	5–12 per 1000 births	≥10%
Homozygous	1–4%	~17%

- **Diagnose by DNA analysis in pregnancy** — the APC-resistance bioassay is unreliable (normal APC resistance develops in pregnancy). **APS can also cause APC resistance**.
- 491 carriers vs 1055 controls: all 3 VTE in carriers, but no difference in preterm birth, birthweight or hypertension.
- **MFMU (~5000 women): 2.7% heterozygous**; none of the VTE (0.8/1000) in carriers; no ↑ preeclampsia, abruption, FGR or loss → **universal screening and prophylaxis for carriers without prior VTE not indicated**.

Prothrombin G20210A mutation

- Missense mutation → excess prothrombin: **+30% heterozygotes, +70% homozygotes**.
- Heterozygote **with** personal/first-degree family history (<50 y): **>10%**; **without: <1%**.
- ~4200 women: **3.8%** carriers (1 homozygous); no ↑ loss, preeclampsia, FGR or abruption.
- **Homozygous or double heterozygous (with FVL) → higher risk**.

Hyperhomocysteinemia

- Commonest cause: **MTHFR C677T thermolabile mutation** (autosomal recessive; printed as "C677T"). Also enzyme defects and deficiency of **folate, B6, B12**.
- Homocysteine **falls in normal pregnancy**; folate supplementation may explain the lack of VTE association in North America.
- **⚠️ ACOG: do not test MTHFR polymorphisms or fasting homocysteine** for VTE evaluation.

Others: low **protein Z** and **PAI-1** promoter polymorphisms — slight ↑ risk; **screening not recommended**.

Acquired thrombophilias

- **Antiphospholipid syndrome (APS), heparin-induced thrombocytopenia, cancer**.

Antiphospholipid syndrome

- Venous and arterial thrombosis: most often **deep leg veins** and **cerebral arteries**.

- **Obstetric criteria** (any one):
 1. **≥1 unexplained fetal death ≥10 wk**
 2. **≥1 preterm birth <34 wk** from eclampsia, severe preeclampsia or placental insufficiency
 3. **≥3 unexplained consecutive spontaneous abortions <10 wk**
- **Test for aPL** with unexplained arterial or venous thrombosis, **one fetal loss**, or **≥3 recurrent embryonic/fetal losses**. Preterm severe preeclampsia or early placental insufficiency alone do not justify testing (aPL may worsen HELLP prognosis).
- **Test all three:** (1) **lupus anticoagulant**, (2) **anticardiolipin IgG and IgM**, (3) **anti-β₂ glycoprotein-I IgG and IgM**. **Repeat any positive test after 12 weeks.**
- **Anticardiolipin = most common sole antibody.** Thrombosis risk **5–12%** in pregnancy/puerperium; up to **25%** of APS thromboses occur then.

Thrombophilias and pregnancy complications

- **⚠ No definite causal link between inherited thrombophilias and adverse pregnancy outcomes.**
- ACOG: **do not screen for inherited thrombophilias** after fetal loss, FGR, abruption or preeclampsia; **no benefit of aspirin/heparin** (unlike APS).
- Stillbirth not ↑ with heterozygous mutations (weak link with **homozygous FVL**). No ↑ FGR. Abruptio: prospective cohorts negative (one older study positive). Preeclampsia: meta-analysis of ~22,000 negative for FVL/prothrombin; 12 thrombophilias negative.
- In contrast, **APS is strongly linked** to fetal loss, recurrent loss and preeclampsia.

Thrombophilia screening

- **Universal screening is not cost-effective** → selective.
- **ACOG — consider screening if:**
 1. **Personal history of VTE** (with or without a recurrent risk factor such as fracture, surgery, prolonged immobilization)
 2. **First-degree relative** (parent or sibling) **with a high-risk inherited thrombophilia**
- **Not** for recurrent fetal loss or adverse outcomes; **no homocysteine screening**.
- Test **≥6 weeks after the thrombosis**, when **not pregnant**, **not anticoagulated**, and **not on hormonal therapy** where possible.

Table 55-3. How to Test for Thrombophilias

Thrombophilia	Testing Method	Reliable During Pregnancy?	Reliable During Acute Thrombosis?	Reliable with Anticoagulation?
Factor V Leiden mutation	Activated protein C resistance assay (second generation)	Yes	Yes	No
	If abnormal: DNA analysis	Yes	Yes	Yes
Prothrombin gene mutation G20210A	DNA analysis	Yes	Yes	Yes
Protein C deficiency	Protein C activity (<65%)	Yes	No	No
Protein S deficiency	Functional assay (<55%)	No^a	No	No
Antithrombin deficiency	Antithrombin activity (<60%)	Yes	No	No

^aIf screening in pregnancy is necessary, cutoff values for free protein S antigen levels in the second and third trimesters have been identified at less than 30% and less than 24%, respectively. From ACOG Practice Bulletin No. 197: Inherited Thrombophilias in Pregnancy, 2018.

Memory hook: "DNA tests never lie; protein tests lie during clots and heparin." Genetic tests (FVL, prothrombin) are reliable anytime; functional protein assays (C, S, AT) are not reliable during acute thrombosis or anticoagulation, and protein S not in pregnancy.

4. Deep-Vein Thrombosis

Clinical presentation

- Pregnancy DVT is mostly **iliofemoral/iliac, without calf involvement** (vs **>80% calf** in the general population).
- **Vast majority left-sided** — the **right iliac artery and the ovarian artery cross the left iliac vein only** (contrast: the **ureter** is compressed more on the right).
- Abrupt **pain and edema of leg and thigh**; reflex arterial spasm may give a **pale, cool limb with ↓ pulses**; calf circumference discordance. Some large clots are nearly silent.
- **Homans sign** (calf pain on squeezing or Achilles stretch) is nonspecific (muscle strain, contusion).
- **⚠️ 30–60% of confirmed DVTs have an asymptomatic PE.**

Diagnosis

- **Clinical diagnosis confirmed in only 10%** → objective testing needed. Most tests are not validated in pregnancy.

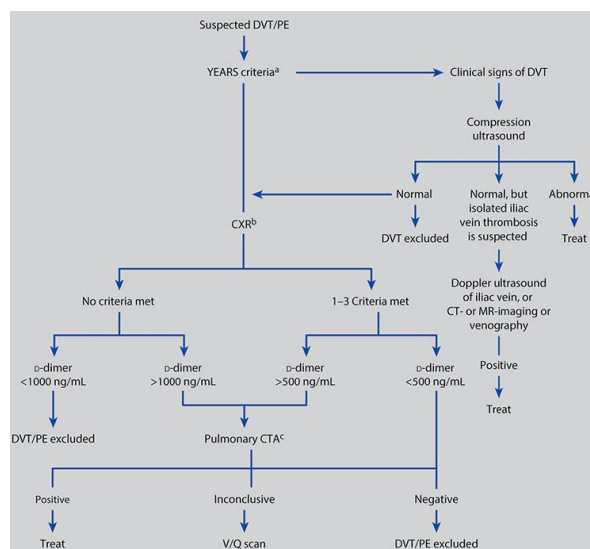


Figure 55-2. Pregnancy-adapted YEARS algorithm: DVT signs → compression ultrasound; otherwise CXR, then D-dimer cutoffs of 1000 ng/mL (no criteria) or 500 ng/mL (1–3 criteria) decide on CT angiography.

Compression ultrasonography (first-line)

- **Initial test for suspected DVT = compression US of the proximal leg veins** (noncompressibility + echo architecture).
- Nonpregnant: normal US → safe to withhold anticoagulation for 1 week; **serial studies** because **~¼ of missed calf clots extend proximally within 1–2 weeks**.
- Pregnancy caveat: normal US does not exclude PE (already embolized, or **iliac/pelvic clot** out of view).
- Two studies (221 and 210 women): **a single negative complete compression US safely excludes DVT in most pregnant women**.
- Isolated **iliac vein thrombosis** suspected → **Doppler of iliac vein, CT/MR, or venography**.

MR imaging

- Excellent above the inguinal ligament → **iliofemoral and pelvic vein thrombosis**; MR venography.

- **Sensitivity 100%, specificity 90%** (nonpregnant). ~½ of those without DVT had another diagnosis (cellulitis, myositis, edema, hematoma, superficial phlebitis).

D-dimer

- Rises with gestational age, multifetal pregnancy, cesarean, abruption, preeclampsia, sepsis → problematic.
- **Pregnancy-adapted YEARS** (3 criteria: **clinical signs of DVT, hemoptysis, PE most likely diagnosis**): PE excluded if **0 criteria + D-dimer <1000 ng/mL**, or **≥1 criterion + D-dimer <500 ng/mL**.
- **DiPEP**: neither clinical risk factors nor D-dimer identified PE accurately → image all.
- ⚠ **ACOG recommends against D-dimer to diagnose VTE in pregnancy** (Parkland agrees), though a **negative D-dimer is reassuring** unless VTE is the most likely diagnosis.

Management of VTE

- **Anticoagulation + limited activity. Thrombophilia testing does not change treatment.**
- **UFH or LMWH** acceptable; **LMWH preferred** (ASH): better bioavailability, longer half-life, predictable response, **less osteoporosis and thrombocytopenia**, less frequent dosing.

Table 55-4. Anticoagulation Regimen Definitions

Anticoagulation Regimen	Definition
Prophylactic LMWH^a	Enoxaparin 40 mg SC once daily ; dalteparin 5000 units SC once daily; nadroparin 2850 units SC once daily; tinzaparin 4500 units SC once daily
Therapeutic LMWH^b	Enoxaparin 1 mg/kg every 12 hours ; dalteparin 200 units/kg once daily; tinzaparin 175 units/kg once daily; dalteparin 100 units/kg every 12 hours. Target anti-Xa 0.6–1.0 units/mL 4 hours after last injection for twice-daily regimen; slightly higher doses may be needed for a once-daily regimen
Prophylactic UFH	5000–7000 units SC q12h (1st trimester); 7500–10,000 units SC q12h (2nd trimester); 10,000 units SC q12h (3rd trimester) unless the aPTT is elevated
Adjusted-dose (therapeutic) UFH^b	≥10,000 units SC q12h , adjusted to target aPTT 1.5–2.5 × control 6 hours after injection
Postpartum anticoagulation	Prophylactic, intermediate or adjusted-dose LMWH for 6–8 weeks as indicated. Oral anticoagulants may be considered postpartum based on planned duration, lactation and patient preference
Surveillance	Clinical vigilance and objective investigation of women with symptoms suspicious of DVT or PE. VTE risk assessment prepregnancy or early in pregnancy , repeated if complications develop (especially hospitalization or prolonged immobility)

^aAt extremes of body weight, modification of dose may be required. ^bAlso referred to as weight-adjusted, full treatment dose. aPTT = activated partial thromboplastin time; LMWH = low-molecular-weight heparin; SC = subcutaneously; UFH = unfractionated heparin. From ACOG Practice Bulletin No. 197, 2018. (An "intermediate" regimen is referred to elsewhere but not defined in this table.)

Duration

- Nonpregnant: **≥3 months**.
- **ACOG: therapeutic anticoagulation for 3–6 months, then intermediate or prophylactic dose for the rest of pregnancy and postpartum.** DOACs can be considered if not breastfeeding.
- Untreated DVT → **PE in up to 60%**; anticoagulation ↓ this to **<5%**. PE mortality (nonpregnant) **~1%**.
- **Recovery**: leg pain settles over days → graded ambulation with elastic stockings after **7–10 days**.
- **Graduated compression stockings for 2 years** ↓ **postthrombotic syndrome** (chronic paresthesia/pain, intractable edema, skin changes, ulcers). Catheter-directed thrombolysis does **not** prevent it (ASH).

Unfractionated heparin

- Consider for **initial treatment** and when **delivery, surgery or thrombolysis** may be needed.
- Two options: IV then adjusted-dose SC q12h, **or** adjusted-dose SC q12h with aPTT checked **6 h post injection**.

- **IV regimen:** bolus **70–100 U/kg (5000–10,000 U)** → infusion **1000 U/h (15–20 U/kg/h)** → titrate to **aPTT 1.5–2.5× control**; IV for **5–7 days**, then SC maintaining aPTT 1.5–2.5×.
- **Lupus anticoagulant** → **aPTT unreliable** → **use anti-Xa levels**.

Low-molecular-weight heparin

- MW **4000–5000 Da** (vs 12,000–16,000 for UFH). **No heparin crosses the placenta**; all act via **antithrombin**.
- UFH: equal anti-Xa and anti-thrombin activity; **LMWH: more anti-Xa**. More predictable, fewer bleeds (bioavailability, longer half-life, dose-independent clearance, less platelet interference).
- **Longer half-life prohibits neuraxial analgesia** in labor. **Renally cleared** → caution with renal dysfunction. More effective than UFH at shrinking thrombus.
- **Monitoring controversial:** ACOG → **anti-Xa 4 h after injection** with dose adjustment; ASH → routine anti-Xa monitoring hard to justify.
- **Clearance ↑ as pregnancy advances** (Fig. 55-3). Tinzaparin **75–175 U/kg/d** needed for peak anti-Xa 0.1–1.0 U/mL; dalteparin **100 U/kg q12h may be insufficient** → slightly higher doses may be needed.
- **No teratogenicity** (enoxaparin). Tinzaparin (1267 women): no HIT, no maternal deaths, no regional-anesthesia complications; bleeding needing intervention **3.4%**.
- **LMWH safe in breastfeeding**.

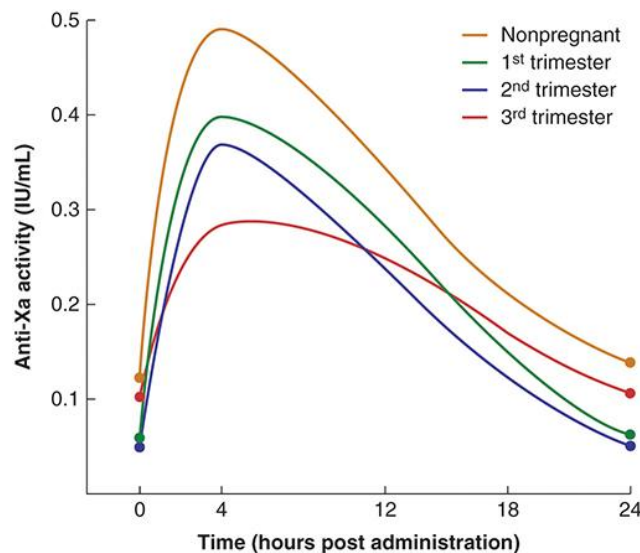


Figure 55-3. Enoxaparin 40 mg daily: peak anti-Xa (at ~4 h) falls from ~0.49 IU/mL nonpregnant to ~0.29 IU/mL in the third trimester (faster clearance).

Labor and delivery (neuraxial timing)

- **Switch LMWH** → **UFH in the last month** (or sooner if delivery imminent) — mainly to avoid **epidural/spinal hematoma**, not delivery bleeding.

Dose	Withhold UFH before neuraxial	Withhold LMWH before neuraxial
Low-dose (prophylactic)	4–6 h	12 h
Intermediate	12 h	12–24 h
Therapeutic	24 h	24 h

Society for Obstetric Anesthesia and Perinatology (Leffert, 2018).

- In labor on UFH → verify clearance with **aPTT**.
- **Protamine sulfate: 1 mg per 100 U heparin, max 50 mg** — rarely needed; **not indicated for prophylactic doses**.
- Anticoagulation paused → **pneumatic compression devices**.

Warfarin (vitamin K antagonists)

- ⚠️ **Contraindicated antepartum** (crosses placenta → fetal death, malformations, hemorrhage) — **sole antepartum use: mechanical heart valve**.

- **Safe in breastfeeding** (doesn't accumulate in milk).
- **Postpartum:** IV heparin + oral warfarin started together; warfarin **5–10 mg for 2 days**, then titrate to **INR 2–3**.
- **Overlap heparin ≥ 5 days AND until INR therapeutic for 2 consecutive days** (avoid paradoxical thrombosis and **skin necrosis** from early anti-protein C effect).
- Puerperal women need **more warfarin (median 45 vs 24 mg)** and **longer (7 vs 4 days)** to reach target INR.

Direct-acting oral anticoagulants (DOACs)

- **Dabigatran** (thrombin inhibitor); **rivaroxaban, apixaban** (Xa inhibitors) — drugs of choice outside pregnancy.
- Reproductive risks essentially unknown; **dabigatran crosses the placenta**; milk excretion unknown → **avoid breastfeeding or use another agent (e.g., warfarin)**.

Complications of anticoagulation

- **Hemorrhage** (most serious; recent surgery/lacerations, excess dose), **thrombocytopenia** and **osteoporosis** (the last two unique to heparin, less with LMWH).

Heparin-induced thrombocytopenia (HIT)

	Type 1 (common)	HIT (severe, immune)
Mechanism	Nonimmune, benign	IgG against platelet factor 4–heparin complexes
Timing	First few days	Days 5–10
Course	Resolves in ~5 days without stopping heparin	Platelets fall >50% (over 1–3 days, from the post-heparin peak) and/or thrombosis ; nadir 40,000–80,000/μL

- Incidence **3–5% nonpregnant vs <0.1% obstetric** (0/244 gravidas vs 10/244 nonpregnant).
- ACCP: **no platelet monitoring if HIT risk <1%**; otherwise **every 2–3 days from day 4 to 14**.
- Management by clinical findings (antibody/functional assays take days): **stop heparin, start alternative** — LMWH may cross-react → **fondaparinux** (pentasaccharide Xa inhibitor; used in pregnancy). **Avoid platelet transfusion.**

Heparin-induced osteoporosis

- With **≥ 6 months** of heparin; worse in smokers; less with LMWH. **Calcium 1500 mg/d** for women on any heparin.

Anticoagulation and abortion

- Heparin does not preclude **careful curettage**; restart full-dose heparin **several hours** after atraumatic evacuation.

Anticoagulation and delivery

- Blood loss depends on (1) dose, route and timing; (2) number/depth of incisions and lacerations; (3) myometrial contraction; (4) other coagulation defects. Heparin is generally **stopped for labor and delivery**.
- **ACOG: restart UFH/LMWH no sooner than 4–6 h after vaginal delivery or 6–12 h after cesarean.** Prophylaxis within 24 h of cesarean appears safe. **Parkland: wait ≥ 24 h** before therapeutic heparin after cesarean or vaginal delivery with significant lacerations.

5. Superficial Venous Thrombophlebitis

- Saphenous system; with **varicosities** or after an **IV catheter**. Treat with **analgesia, elastic support, heat, rest**.
- **$\uparrow$ DVT risk 4–6 $\times$** ; meta-analysis: **18% concomitant DVT, 7% PE**. ASH suggests **LMWH** (controversial).

6. Pulmonary Embolism

- **$\sim 10\%$ of maternal deaths. 70%** of gravidas with PE have clinical DVT; **30–60%** of DVTs have silent PE.

Clinical presentation

- (~2500 nonpregnant patients) **dyspnea 82%, chest pain 49%, cough 20%, syncope 14%, hemoptysis 7%**.
- Tachypnea, apprehension, tachycardia; $\pm$ loud P2, rales, friction rub.
- **ECG**: right-axis deviation, anterior T-wave inversion. **CXR usually normal** (or atelectasis, infiltrate, cardiomegaly, effusion, regional oligemia).
- ⚠️ **Normal ABG does not exclude PE**: $\sim 1/3$ of young patients have $PO_2 > 80$ mm Hg. **A-a gradient > 20 mm Hg in $> 86\%$** — more useful.

Massive and submassive PE

- **Massive = hemodynamic instability, 5–10%** of cases. Obstruction $\rightarrow \uparrow$ PVR, pulmonary hypertension, **acute RV dilation**.
- Significant pulmonary hypertension needs **60–75% occlusion**; **circulatory collapse needs 75–80%** $\rightarrow$ acutely symptomatic emboli are usually large (**saddle**).
- **Submassive = RV dysfunction**; mortality **$\sim 25\%$ vs 1%** without RV dysfunction.
- ⚠️ **Give crystalloid carefully; support BP with vasopressors** — aggressive fluids worsen RV dysfunction. O_2 , intubation and ventilation in preparation for thrombolysis, filter or embolectomy.

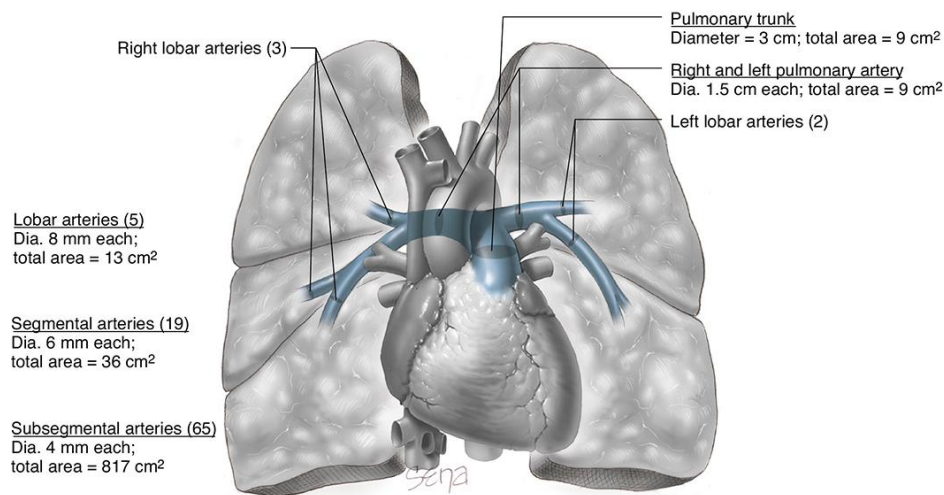


Figure 55-4. Pulmonary arterial cross-sectional area: trunk/main arteries 9 cm², 5 lobar 13 cm², 19 segmental 36 cm², 65 subsegmental 817 cm² — a saddle embolus (50–90% occlusion) causes instability; emboli beyond the lobar arteries rarely do. (The printed legend gives these areas in "cm", meaning cm².)

Diagnosis

- Needs **high index of suspicion**. Radiation concern is outweighed by the danger of missing PE; a false diagnosis is also harmful (anticoagulation, delivery plans, contraception, future prophylaxis) $\rightarrow$ aim for **diagnostic certainty**.
- Same algorithm as DVT (Fig. 55-2): **leg compression US**, CXR and ECG; **echo** to exclude mimics (MI, tamponade, aortic dissection). Confirm with **CTPA, MR or V/Q scan**.
- **ASH: V/Q scan first-line; Parkland: CTPA preferred**.

Test	Fetal dose	Maternal breast dose	Notes
CTPA (multidetector)	0.45–0.6 mGy	10–70 mGy	Most used; detects small distal emboli of uncertain significance; negative CTPA → no VTE at follow-up (318 women)
V/Q scan (^{99m} Tc-MAA)	0.1–0.4 mGy (negligible, both)	negligible	¼ nondiagnostic in pregnancy (pneumonia, bronchospasm) → then CTPA
Pulmonary angiography	Higher	—	Reference standard but invasive; reserved for equivocal cases

- CTPA vs V/Q in 137 pregnant women: comparable; **~20% indeterminate** with both. Either is appropriate to exclude PE.

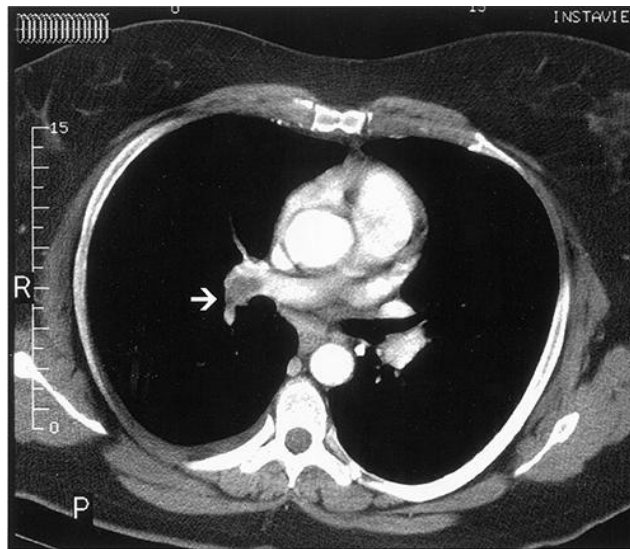


Figure 55-5. CT pulmonary angiogram: large filling defect (thrombus) in the right pulmonary artery (arrow).

Management

- **Full anticoagulation** as for DVT (Table 55-4), plus as needed:

Vena caval filters

- **Recent PE + need for cesarean** → consider filter **before surgery** (reversal risks re-embolism; full anticoagulation risks hemorrhage).
- Also: **recurrent PE despite heparin**, or massive PE when thrombolysis isn't possible.
- Jugular or femoral insertion, even in labor. **Routine filters add nothing** to heparin. **Retrievable filters** removed at **1–2 wk**. Pregnancy complication rates ≈ nonpregnant; one series **29%** complications (incl. removal failure).

Thrombolysis

- Faster clot lysis and ↓ pulmonary hypertension in life-threatening PE. Submassive PE: **alteplase** ↓ death or escalation.
- Meta-analysis: recurrence or death **10% vs 17%** (lytic + heparin vs heparin), but **2% fatal bleeding** with lysis.
- Pregnancy: complications similar to nonpregnant → **don't withhold if indicated**. Within 48 h of delivery: transfusion in 5/8 cesareans (3 hysterectomies). 141 pregnant women: **maternal death 3%, major bleeding 9%**.

Embolectomy

- Rarely needed. Maternal operative risk reasonable, but **stillbirth 20–40%**.

7. Thromboprophylaxis

- **No consensus** (Cochrane) on who should receive prophylaxis; ASH mostly "suggests" rather than "recommends". Unindicated intervention is common.

- Decision depends on **personal and family VTE history, thrombophilia severity, and extra risk factors** (obesity, cesarean, immobility).

Table 55-5. Some Recommendations for Thromboprophylaxis During Pregnancy

Clinical Scenario	Pregnancy — ACOG ^a	Pregnancy — ASH ^b	Postpartum — ACOG ^a	Postpartum — ASH ^b
Prior single VTE				
Risk factor no longer present	Surveillance only	Surveillance only	Surveillance <i>or</i> prophylactic heparin with additional risk factors ^f	Prophylactic <i>or</i> intermediate-dose heparin
Pregnancy- or estrogen-related, or no known association	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic, intermediate <i>or</i> therapeutic-dose heparin	Prophylactic <i>or</i> intermediate-dose heparin
Associated with a high-risk thrombophilia ^d or affected first-degree relative	Prophylactic, intermediate <i>or</i> therapeutic-dose heparin	NSS	Prophylactic, intermediate <i>or</i> therapeutic-dose heparin	Prophylactic <i>or</i> intermediate-dose heparin
Associated with a low-risk thrombophilia ^e	Prophylactic <i>or</i> intermediate-dose heparin	NSS	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic <i>or</i> intermediate-dose heparin
Two or more prior VTEs ± thrombophilia				
Not receiving long-term anticoagulation	Intermediate <i>or</i> therapeutic-dose heparin	NSS	Intermediate <i>or</i> therapeutic-dose heparin	Prophylactic <i>or</i> intermediate-dose heparin
Receiving long-term anticoagulation	Therapeutic-dose heparin	Therapeutic-dose heparin	Resume long-term anticoagulation	Resume long-term anticoagulation
No prior VTE				
No thrombophilia	Surveillance	NSS	Surveillance <i>or</i> prophylactic heparin with additional risk factors ^f	NSS
High-risk thrombophilia	Surveillance only <i>or</i> prophylactic-dose heparin	Surveillance	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic heparin
Positive family history VTE <i>and</i> low-risk thrombophilia ^c	Surveillance <i>or</i> prophylactic-dose heparin	Surveillance	Prophylactic <i>or</i> intermediate-dose heparin	Surveillance
Low-risk thrombophilia	Surveillance	Surveillance	Surveillance; prophylactic-dose heparin with additional risk factors ^f	Surveillance
Compound heterozygous thrombophilia	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic heparin	Prophylactic <i>or</i> intermediate-dose heparin	Prophylactic heparin
Antiphospholipid antibodies				
History of VTE	Prophylactic heparin (? plus low-dose aspirin)	NSS	Prophylactic heparin; referral to specialist ^g	NSS
No prior VTE	Surveillance <i>or</i> prophylactic heparin plus low-dose aspirin with recurrent pregnancy loss or stillbirth ^h	NSS	Prophylactic heparin plus low-dose aspirin × 6 weeks with recurrent pregnancy loss or stillbirth ^g	NSS

^aACOG 2020a,b. ^bAmerican Society of Hematology (Bates, 2018). ^cPostpartum treatment levels should be $\geq$ antepartum treatment. ^dHigh-risk thrombophilia: antithrombin deficiency; doubly heterozygous or homozygous for prothrombin 20210A and factor V Leiden. ^eLow-risk thrombophilia: heterozygous factor V Leiden or prothrombin 20210A; protein S or C deficiency. ^fFirst-degree relative with VTE at <50 years; other major thrombotic risk factors, e.g., obesity, prolonged immobility. ^gWomen with antiphospholipid syndrome should not use estrogen-containing contraceptives. ^hTreatment is recommended if the diagnosis of APS is based on three or more prior pregnancy losses. NSS = not specifically stated; VTE = venous thromboembolism. Regimens are in Table 55-4. In the printed table, footnote h is never referenced and the APS "No prior VTE" pregnancy cell carries f; h is clearly the intended footnote and is placed there above.

Prior VTE

- Generally **antepartum surveillance or heparin prophylaxis** for prior VTE without a recurrent risk factor.
- Evidence:
 - 87 Swedish women (not tested for thrombophilia): antepartum recurrence **15% despite UFH 5000 U bid vs 12% without** → low-dose UFH may be ineffective.
 - 125 women, single prior VTE, no antepartum heparin (4–6 wk postpartum only): 6 recurrences; **none if no thrombophilia or the prior clot had a temporary risk factor**.
 - 88 women without aPL, no prophylaxis: **22%** pregnancy/puerperal VTE; transient-risk-factor group (not pregnancy/OC): no antepartum recurrences, 2 postpartum.
- → **Prior VTE with thrombophilia or without a temporary risk factor** → **antepartum and postpartum prophylaxis**. Tailor to the circumstances of the original event.
- VTE can recur despite prophylaxis: **10%** of 270 women (12 before prophylaxis started, 16 on LMWH).
- **Parkland**: formerly UFH 5000–7500 U SC 2–3× daily; now **enoxaparin 40 mg SC daily**, switched to UFH near delivery.

Cesarean delivery

- DVT and especially **fatal PE risk rises many-fold** after cesarean; 1/3 of US births are cesarean.
- Guidelines vary hugely — proportion of a cohort qualifying for pharmacologic prophylaxis: **ACOG ~1%, ACCP ~35%, RCOG ~85%** (SMFM).

Table 55-6. Thromboprophylaxis Guidelines for Women Undergoing Cesarean Delivery

Organization	Recommendations	Risk Factors
ACOG ^a	Pneumatic compression devices for all ; pharmacological prophylaxis for risk factors	Risk factors not defined; individualize therapy
SMFM ^b	Pneumatic compression devices for all; prophylactic LMWH; intermediate-dose LMWH; individualized therapy	History of VTE; thrombophilia without history of VTE; class III obesity with a thrombophilia or history of VTE ; combination of the above factors
RCOG ^c	Prophylactic LMWH for 10 days	Cesarean during labor ; BMI >40 kg/m² ; medical comorbidities; or ≥2 of : age >35, parity >3, obesity, smoker, varicose veins, systemic infection, preeclampsia, preterm delivery, stillbirth, operative vaginal delivery, PPH, prolonged admission
RCOG	Prophylactic LMWH for 6 weeks	Prior VTE; high-risk thrombophilia; low-risk thrombophilia and family history of VTE; antenatal LMWH therapy

^aACOG 2020b. ^bSMFM 2020. ^cRCOG 2015. BMI = body mass index; LMWH = low-molecular-weight heparin; PPH = postpartum hemorrhage; VTE = venous thromboembolism.

- **ACOG (since 2011): pneumatic compression devices before cesarean for all** women not already on prophylaxis; add UFH/LMWH for additional risk factors. **Don't delay an emergency cesarean** to start prophylaxis.
- HCA implementation: PE deaths fell from **7/458,097 to 1/465,880** cesareans.

- National Partnership for Maternal Safety (2016): expand prophylaxis to antenatal admissions **≥3 days**, vaginal births, and most cesareans — criticized as based on sparse data (Parkland agrees with the critics).

Quick-Recall: Thrombophilias at a Glance

Thrombophilia	Inheritance / prevalence	Mechanism	Key facts
Antithrombin deficiency	AD; ~1/2500; homozygous lethal	↓ thrombin/Xa inactivation	Most thrombogenic ; 40% VTE with prior VTE; anticoagulate all; recombinant AT
Protein C deficiency	AD; 2–3/1000	↓ inactivation of Va/VIIIa	6–12x; homozygous → neonatal purpura fulminans
Protein S deficiency	0.3–1.3/1000	Cofactor for protein C	Levels fall in pregnancy (free S <30% / <24% cutoffs)
Factor V Leiden	Most common ; 3–15% Europeans	APC resistance (Arg506Gln)	~40% of pregnancy VTE ; DNA test in pregnancy
Prothrombin G20210A	3.8%	↑ Prothrombin (30% / 70%)	<1% without history; >10% with history
Hyperhomocysteinemia (MTHFR)	AR	Inhibits protein C activation	Don't test
Antiphospholipid syndrome	Acquired	Venous + arterial	LA, aCL, anti-β ₂ GPI; repeat at 12 wk; strong link to loss/preeclampsia

Key Numbers to Memorize

- VTE **1–2 per 1000** pregnancies; **5x** nonpregnant; PE **~10%** of pregnancy-related deaths
- Postpartum VTE **22 / 100,000** (0–6 wk) → **3 / 100,000** (7–12 wk)
- Leg venous flow ↓ **50%** (early 3rd trimester → 6 wk postpartum)
- Prior VTE = **15–25%** of pregnancy VTE; thrombophilia in **20–50%**
- Antithrombin: heparin accelerates **>1000x**; RR **25–50x**
- FVL heterozygous: **5–12/1000** without history, **≥10%** with; homozygous **1–4% / 17%**
- APS: fetal death **≥10 wk**, preterm **<34 wk**, **≥3** losses **<10 wk**; repeat test **12 wk**; thrombosis **5–12%**
- Test thrombophilia **≥6 wk** after the event, off anticoagulation/hormones, not pregnant
- DVT: **left-sided**, iliofemoral; **30–60%** silent PE; clinical diagnosis confirmed in **10%**
- YEARS: D-dimer **<1000** (0 criteria) / **<500** ng/mL (≥1 criterion)
- UFH IV: bolus **70–100 U/kg**, **1000 U/h** (15–20 U/kg/h), aPTT **1.5–2.5x**
- Enoxaparin **40 mg daily** (prophylactic) / **1 mg/kg q12h** (therapeutic); anti-Xa **0.6–1.0** at **4 h**
- Therapeutic anticoagulation **3–6 months**, then prophylactic through **6–8 wk postpartum**
- Neuraxial: stop UFH **4–6 / 12 / 24 h**, LMWH **12 / 12–24 / 24 h**; switch to UFH in the **last month**
- Restart heparin **4–6 h** after vaginal, **6–12 h** after cesarean (Parkland ≥24 h)
- Protamine **1 mg / 100 U**, max **50 mg**
- Warfarin **5–10 mg × 2 days**, INR **2–3**; overlap **≥5 days** + INR ×2 days
- HIT: day **5–10**, platelets ↓ **>50%**, nadir **40–80 × 10³/μL**; **<0.1%** obstetric → **fondaparinux**
- Heparin osteoporosis: **≥6 months**; calcium **1500 mg/d**
- PE: dyspnea **82%**; A–a gradient **>20 mm Hg** in **>86%**; collapse needs **75–80%** occlusion; submassive mortality **25%**
- CTPA fetal **0.45–0.6 mGy** (breast 10–70); V/Q **0.1–0.4 mGy**; V/Q nondiagnostic **¼**
- Thrombolysis in pregnancy: maternal death **3%**, major bleeding **9%**; embolectomy stillbirth **20–40%**
- Cesarean prophylaxis eligibility: ACOG **1%**, ACCP **35%**, RCOG **85%**; RCOG LMWH **10 days** or **6 weeks**